

Excel Data Tools

Lesson 27 — Sort, Filter, Charts and PivotTables

JKSSB Computer Awareness

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What Excel Data Tools Do

- ▶ Data entered in a worksheet must be **organised**, **summarised**, and **presented**.
- ▶ **Sort** and **filter** arrange and narrow the rows.
- ▶ **Tables**, **charts**, and **PivotTables** organise and summarise the data.
- ▶ **Printing**, **protection**, and **what-if** analysis prepare and safeguard the result.

The one idea

These tools change the *view* of the data; they do not change the data itself.

Sorting

- ▶ **Sort** rearranges rows into a chosen order.
- ▶ **Ascending** order: A to Z, or smallest number to largest.
- ▶ **Descending** order: Z to A, or largest to smallest.
- ▶ Sorting works on a **column**, while the whole row moves with it.

Data > Sort

The Sort dialog can order by **more than one level** — for example, by Department, and then by Salary within each department.

Multi-Level and Custom Sort

- ▶ **Add Level** in the Sort dialog sorts by a second and a third key.
- ▶ A **custom list** sorts by a defined order, such as days of the week or months.
- ▶ If the range has titles, tick **My data has headers** so the header row is not sorted as data.
- ▶ Excel can also sort by cell colour or font colour.

Trap alert

Sorting a **single column** on its own scrambles the records. Sort the whole table, or use an Excel table.

Filtering with AutoFilter

- ▶ **Filter** shows only the rows that meet a condition; the other rows are **temporarily hidden**, not deleted.
- ▶ Turn it on with **Data > Filter**, shortcut **Ctrl+Shift+L**.
- ▶ Drop-down arrows appear in the header row of each column.
- ▶ Clicking an arrow narrows the list to the chosen values.

Hidden, not removed

Clearing the filter (**Clear Filter**) brings back every row.

Filter Criteria

- ▶ **Value filter**: keep one or more ticks from the list.
- ▶ **Condition filter**: Text Filters, Number Filters, and Date Filters (for example, Greater Than 500, or Begins With “A”).
- ▶ **Top 10** and **Above Average** are ready-made condition filters.
- ▶ **Advanced Filter** handles complex criteria and can copy the result to another location.

Excel Tables (Format as Table)

- ▶ An **Excel table** is a formatted range with a name and a table style.
- ▶ Create one with **Insert > Table**, shortcut **Ctrl+T**.
- ▶ A table auto-expands as new rows are typed, and formulas auto-fill.
- ▶ It provides banded rows, header filter buttons, and an optional **Total Row**.

Structured references

In a table, formulas use names such as **Table1[Sales]** instead of cell addresses like **B2**, so they stay correct as the table grows.

Table vs Plain Range

Point	Plain range	Excel table
Name	None	Assigned name (Table1)
Grows	Manually	Auto-expands
Formulas	Cell addresses	Structured references
Filters	Manual AutoFilter	Built-in header buttons

A table is not a PivotTable; a table is the source data that a PivotTable can summarise.

Charts: Common Types

- ▶ **Column** or **bar** chart: compares values across categories.
- ▶ **Line** chart: shows a trend, usually over time.
- ▶ **Pie** or **doughnut** chart: shows parts of one whole.
- ▶ **Scatter (XY)** chart: shows the relationship between two numeric variables.

Linked to data

A chart is **linked** to its source range; edit the data and the chart updates.

Chart Parts

- ▶ **Chart area** is the whole chart; **plot area** is where the data itself is drawn.
- ▶ A **data series** is one set of values; the **legend** names each series.
- ▶ The **category axis** (X) holds labels; the **value axis** (Y) holds numbers.
- ▶ A chart can float on a worksheet or be moved to a separate **chart sheet**.

Data labels print the exact value on each data point.

PivotTables

- ▶ A **PivotTable** summarises a large table of data in seconds.
- ▶ Fields are **dragged** into four areas: **Rows**, **Columns**, **Values**, and **Filters**.
- ▶ The **Values** area aggregates by **Sum**, **Count**, **Average**, and more.
- ▶ Change the field layout and the summary is **re-computed** at once — that is the “pivot”.

Source unchanged

A PivotTable does not edit the source data; it is only a summary view of it.

PivotTable Areas and PivotCharts

Area	What goes there
Rows	Field values listed down the left side
Columns	Field values spread across the top
Values	Numbers that are summarised (Sum, Count, Average)
Filters	Fields used to filter the whole PivotTable

A **PivotChart** is a chart built on a PivotTable; the two are linked.

PivotTable vs Source Data

- ▶ The PivotTable **summarises** the source; it never replaces it.
- ▶ If the source data changes, the PivotTable must be **refreshed**.
- ▶ Deleting the source table breaks the PivotTable.
- ▶ A PivotTable is **not** the same as an Excel table.

Trap alert

A PivotTable “summarises data but is not the source table itself”. Do not confuse the summary with the data.

Printing: Page Setup

- ▶ Print from **File > Print**; adjust layout on the **Page Layout** tab.
- ▶ **Orientation**: **Portrait** (tall) or **Landscape** (wide).
- ▶ Set **paper size**, **margins**, and **headers/footers** here.
- ▶ Preview in **Print Preview** before printing a large sheet.

Keep it on one page

Scaling options such as “Fit Sheet on One Page” shrink the printout to a chosen width or number of pages.

Printing: Area, Titles and Gridlines

- ▶ **Print Area**: select a range and **Set Print Area** to print only that part.
- ▶ **Print Titles** repeats chosen **header rows** or columns on every printed page.
- ▶ **Gridlines** are **not printed by default**; enable **Print** gridlines under Sheet Options.

Page breaks can be inserted to control where a new page starts.

Protection

- ▶ **Protect Sheet** (**Review** > **Protect Sheet**) stops cells from being edited; selected actions may still be allowed.
- ▶ **Protect Workbook** protects the **structure** — adding, deleting, or renaming sheets.
- ▶ Cells are **Locked** by default, but locking takes effect only **after** the sheet is protected.

Not the same as encryption

Protection restricts editing inside Excel; it is not the same as a password that encrypts the file.

What-If Analysis (Awareness)

- ▶ **Goal Seek** finds the input value needed to reach a target result.
- ▶ **Scenario Manager** stores and compares several sets of input values.
- ▶ A **Data Table** shows how an output changes as one or two inputs change.
- ▶ These tools live under **Data > What-If Analysis**.

Exam depth: recognise each tool and its one-line purpose.

Near-Neighbour: COUNTIFS vs SUMPRODUCT

- ▶ To **count rows meeting several conditions at once**, use **COUNTIFS(range1, crit1, range2, crit2, ...)**.
- ▶ **COUNTIFS** applies **AND** logic across all its range-and-criteria pairs.
- ▶ **SUMPRODUCT((range1>=x)*(range2>=y))** also counts multi-criterion rows, by multiplying TRUE/FALSE results.
- ▶ For **OR** logic, add two COUNTIF or COUNTIFS results.

Trap alert (TRAP-14, PYQ-15)

COUNTIFS takes **paired** range-and-criteria arguments; SUMPRODUCT works on **arrays** and adds the already-multiplied boolean results directly.

Trap Alert: Data-Tools Facts to Keep Straight

- ▶ Sort and filter change the **view**; a filter **hides** rows, it does not delete them.
- ▶ A PivotTable **summarises** the source and does not replace it; it must be **refreshed** after changes.
- ▶ A PivotTable is **not** an Excel table.
- ▶ **COUNTIFS** counts with AND across paired arguments; **SUMPRODUCT** does the same with boolean arrays. (TRAP-14)
- ▶ Locked cells are protected only **after** the sheet is protected.
- ▶ Gridlines are **not printed** unless Print gridlines is enabled.

Lesson 27: Recall

- ▶ **Sort** arranges rows — ascending, descending, multi-level, and custom.
- ▶ **Filter** (**Ctrl+Shift+L**) shows only matching rows; it hides, not deletes.
- ▶ An **Excel table** (**Ctrl+T**) auto-expands and uses structured references.
- ▶ Charts (column, line, pie, scatter) are **linked** to their data.
- ▶ A **PivotTable** summarises with Rows, Columns, Values, and Filters; a **PivotChart** is a chart on a PivotTable.
- ▶ Printing: print area, print titles, scaling, and gridline options.
- ▶ **Protect Sheet** vs **Protect Workbook**; locking applies after protection.
- ▶ **Goal Seek**, **Scenario Manager**, and **Data Table** for what-if analysis.
- ▶ **COUNTIFS** vs **SUMPRODUCT** for multi-criterion counting (TRAP-14).